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Nonlinear transport in non-centrosymmetric systems

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Ohm's law has been a cornerstone of electronics since its experimental discovery. This law establishes that, in a conductive system, the voltage is directly proportional to the current. Even when time-reversal symmetry is disrupted, leading to the emergence of magnetoresistance and Hall effects, the linear relationship between voltage and current remains intact. However, recent experiments have demonstrated a breakdown of Ohm's law in non-centrosymmetric structures. In these systems, nonlinear transport effects are permitted with quadratic scaling between voltages and currents. Here we review the main demonstrations of nonlinear transport in non-centrosymmetric systems, analysing the connection between nonlinear behaviour and the system's symmetry. We also investigate the microscopic mechanisms driving these effects, such as Berry curvature dipole and Berry connection polarizability. Finally, we highlight potential applications of nonlinear transport in spintronics and energy harvesting.

In 1827, G. Ohm established that voltage (V) and current (I) are linearly proportional, $V = IR$, where R is the resistance of the electrical conductor[1]. In his experiments, a galvanometer was used to measure the current between the terminals of a thermocouple, in which the voltage scales with the temperature gradient. By experimenting with different materials, wire lengths and temperature gradients, he consistently observed that V and I are parallel and linearly proportional. This empirical observation, known as Ohm's law, remains one of the most important principles in modern electronics. In 1857, W. Thomson discovered that the magnitude of the resistance can be modified via the application of an external magnetic field (B)[2]. Twenty-two years later, E. H. Hall revealed that such magnetic fields, or the presence of magnetization, not only affected the resistance value but also induced a voltage perpendicular to the applied current[3] ('Longitudinal and transverse versus ohmic (dissipative) and Hall (non-dissipative), in Box 1). However, despite the breaking of time-reversal symmetry ($\mathcal{T}$), the linear relationship between voltage and current remains intact. In this case we can write, in a more general fashion, $V = R^{(1,0)}I + R^{(1,2)}IB^2 + R^{(1,1)}IB$, where $R^{(1,0)}$, $R^{(1,2)}$ and $R^{(1,1)}$ refer to the ohmic, magnetoresistance and Hall components, respectively (Fig. 1a). Throughout this Review, we use the notation $R^{(a,b)}$, where a denotes the current and b refers to the magnetic field scaling order.

Non-centrosymmetric systems, where inversion symmetry ($\mathcal{P}$) is broken, do enable nonlinear transport effects. In simple terms, electrons experience different resistance when travelling in opposite directions, as they are not equivalent. In this Review, we focus on second-order transport effects, although higher-order effects are also possible[4–7]. As we will detail below, the second-order effects cause the V–I characteristics of $\mathcal{P}$-broken systems to be non-reciprocal, such that $V(+I) \neq -V(-I)$. In this regard, the V–I characteristics of non-centrosymmetric materials resemble those of classical diodes[8], whereas they differ from other systems, such as tunnel[9,10] or Josephson[11,12] junctions, that exhibit nonlinear yet reciprocal responses. To stress this peculiarity, the second-order nonlinear transport in non-centrosymmetric systems is often referred to as non-reciprocal.

Nonlinear electrical transport effects in non-centrosymmetric systems were first studied in chiral crystals. Previous studies had reported an optical effect, referred to as magnetochiral anisotropy, which depends on both the wave vector of the light and the external magnetic field B (ref. 13). Shortly thereafter, its electrical counterpart—electrical magnetochiral anisotropy—was also observed[14,15]. In addition to the linear components, the voltage has a contribution that scales quadratically with the current and linearly with the external magnetic

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BOX 1

Understanding transport responses—nomenclature and measurement approach

Longitudinal and transverse versus ohmic (dissipative) and Hall (non-dissipative)

In transport experiments, a current (I) is injected and voltage drops are measured in both parallel ($V_{\parallel}$) and transverse ($V_{\perp}$) configurations. These signals are commonly referred to as ohmic and Hall responses, respectively, but, in general, this is not accurate. In linear transport, Onsager's reciprocity states that $\mathcal{T}$-even ($\mathcal{T}$-odd) linear responses are symmetric (antisymmetric) under the exchange of voltage and current[102]. Therefore, $\mathcal{T}$-even responses are associated with dissipative or ohmic processes ($V \cdot I \neq 0$) whereas $\mathcal{T}$-odd responses are related to non-dissipative or Hall processes ($V \cdot I = 0$)[138].

In high-symmetry systems, Hall and transverse are indeed equivalent: the longitudinal linear responses are $\mathcal{T}$-even, $V_{\parallel} = R^{(1,0)}I + R^{(1,2)}IB^2$, whereas the linear transverse responses are $\mathcal{T}$-odd, $V_{\perp} = R^{(1,1)}IB$. However, in lower-symmetry systems, $\mathcal{T}$-even voltages may also emerge in transverse configurations. Symmetric, dissipative transverse responses include off-diagonal resistances ($V_{\perp} = R^{(1,0)}I$)[140] and planar Hall effects ($V_{\perp} = R^{(1,2)}IB^2$)[26,27,141], which are not true Hall effects as they are symmetric under current–voltage exchange. Moreover, even in the absence of magnetic fields or off-diagonal resistances, currents applied off-axis in anisotropic systems will produce a transverse voltage that is strictly related to longitudinal transport components[32]. In conclusion, transverse is generally not synonymous with Hall.

In the nonlinear regime, Onsager's reciprocity does not apply, and antisymmetric responses under the exchange of voltage and current may emerge without $\mathcal{T}$ breaking. However, ohmic and Hall responses are still defined[44] by whether they are dissipative ($V \cdot I \neq 0$) or non-dissipative ($V \cdot I = 0$). As with the linear case, a single measurement of a transverse nonlinear voltage is not sufficient to determine whether it is a Hall voltage ($V \cdot I = 0$)[85]. Therefore, we suggest using the terms longitudinal and transverse to refer to the respective components of the nonlinear transport tensors while reserving ohmic and Hall for actual dissipative and non-dissipative responses, respectively[138].

Disentangling the nonlinear transport response

Nonlinear effects are weaker than their linear counterparts. In DC measurements, the DC voltage (V^{DC}) contains both linear and nonlinear components, making nonlinear contributions difficult to disentangle. A common practice is to perform AC transport measurements at very low frequency (ω) values, which resemble DC characteristics but enable independent measurement of the first-harmonic (V^{ω}) and second-harmonic ($V^{2\omega}$) voltages, containing only linear and second-order contributions, respectively. Thus, by measuring $V^{2\omega}$, it is possible to accurately characterize the nonlinear transport effects. Note that $V^{2\omega}$ depends quadratically on I^{ω} and changes sign when the voltage and current directions are reversed simultaneously. Moreover, whereas the first-harmonic resistance (V^{ω}/I^{ω}) is independent of the current, the second-harmonic resistance ($V^{2\omega}/I^{\omega}$) scales linearly with it.

field (Fig. 1b). By measuring the second-harmonic voltage ($V^{2\omega}$) in response to a current at frequency ω (I^{ω}), it is possible to isolate such a contribution: $V^{2\omega} = R^{(2,1)}(I^{\omega})^2B$ ('Disentangling the nonlinear transport response' in Box 1). Subsequently, the effect was generalized to other non-centrosymmetric systems, such as polar semiconductors[16], topological insulators[17] and two-dimensional electron gases[18]. The nonlinear transport effect in the presence of an external magnetic field may emerge in both longitudinal and transverse configurations. The longitudinal manifestation, where the voltage is parallel to the current ($V_{\parallel}^{2\omega} = R^{(2,1)}(I^{\omega})^2B$), has been named bilinear magnetoelectric resistance[17,19], bilinear magnetoresistance[20,21] or unidirectional magnetoresistance[22–24], as the second-harmonic resistance ($V^{2\omega}/I^{\omega}$) depends linearly on both the current and the magnetic field. The transverse effect, in which the voltage is transverse to the current ($V_{\perp}^{2\omega} = R^{(2,1)}(I^{\omega})^2B$), has also been observed and referred to as the nonlinear planar Hall effect[25], by analogy to its linear counterpart—the planar Hall effect[26,27] ('Longitudinal and transverse versus ohmic (dissipative) and Hall (non-dissipative), in Box 1). In addition, other terms such as non-reciprocal electrical transport[28,29] can also be found in the literature, potentially leading to misunderstandings among readers. We propose to use the term nonlinear magnetoconductivity (NLMC) for any $V^{2\omega}$ that scales quadratically with the current and linearly with the magnetic field ($V^{2\omega} = R^{(2,1)}(I^{\omega})^2B$), independently of the measurement configuration or system studied.

In the absence of an external magnetic field, nonlinear transport effects have also been observed recently. The nonlinear Hall effect (NLHE), which consists of $V^{2\omega}$ purely transverse to the current, was first observed in WTe_2 (refs. 30,31). It immediately attracted much attention because, until that point, Hall effects had been restricted to systems with broken $\mathcal{T}$. However, the Onsager reciprocity theorem does not apply in the nonlinear regime, allowing NLHE in systems with broken $\mathcal{P}$ but preserved $\mathcal{T}$. Later, the longitudinal counterpart ($V_{\parallel}^{2\omega} = R^{(2,0)}(I^{\omega})^2$), known as the nonlinear conductivity (NLC), was also observed[32,33] (Fig. 1c). Both manifestations have been studied intensively in polar[30,31] and chiral[32] systems. Moreover, engineered structures have been developed to break $\mathcal{P}$, yielding strong nonlinear responses[34] which can be exploited for energy harvesting applications[35]. In general, the term 'Hall' should be reserved for non-dissipative responses, but it is often used to describe any transverse voltage (for clarification on nomenclature, see 'Longitudinal and transverse versus ohmic (dissipative) and Hall (non-dissipative) in Box 1). A more precise approach is to use the term 'NLC' to broadly describe the nonlinear transport effect at zero field, with longitudinal and transverse NLC denoting the respective components of the response. This definition leaves the term 'NLHE' for true Hall responses and standardizes the notation, enabling the use of NLC ($V^{2\omega} = R^{(2,0)}(I^{\omega})^2$) and NLMC ($V^{2\omega} = R^{(2,1)}(I^{\omega})^2B$) to distinguish between nonlinear transport effects in the absence and presence, respectively, of an external magnetic field.

In summary, in a centrosymmetric conductor, the voltage response to an applied electrical current has three linear contributions with the following dependence: $V = R^{(1,0)}I + R^{(1,2)}IB^2 + R^{(1,1)}IB$. However, in a non-centrosymmetric conductor, besides these three linear components, two nonlinear contributions emerge. Consequently, the voltage response up to second order in the current follows the equation $V = R^{(1,0)}I + R^{(1,2)}IB^2 + R^{(1,1)}IB + R^{(2,1)}I^2B + R^{(2,0)}I^2$, where $R^{(2,1)}$ and $R^{(2,0)}$ denote the NLMC and the NLC components, respectively (Fig. 1). Although higher-order terms in the magnetic field are theoretically possible, their expected lower magnitude renders experimental detection challenging. Previous reviews have discussed general nonlinear phenomena[36–38] or have focused on transport effects, but only in specific systems, such as van der Waals nanostructures[39], or on a particular effect, such as NLHE[40–42]. However, despite the rapid growth of the field, a comprehensive review that offers a general yet thorough overview of nonlinear transport phenomena in non-centrosymmetric systems, while incorporating the latest contributions, remains lacking. In this Review, the main reports on nonlinear transport effects in non-centrosymmetric systems will be examined, covering polar and chiral systems, as well as magnetic materials and engineered structures. Importantly, given the intimate link between nonlinear effects and

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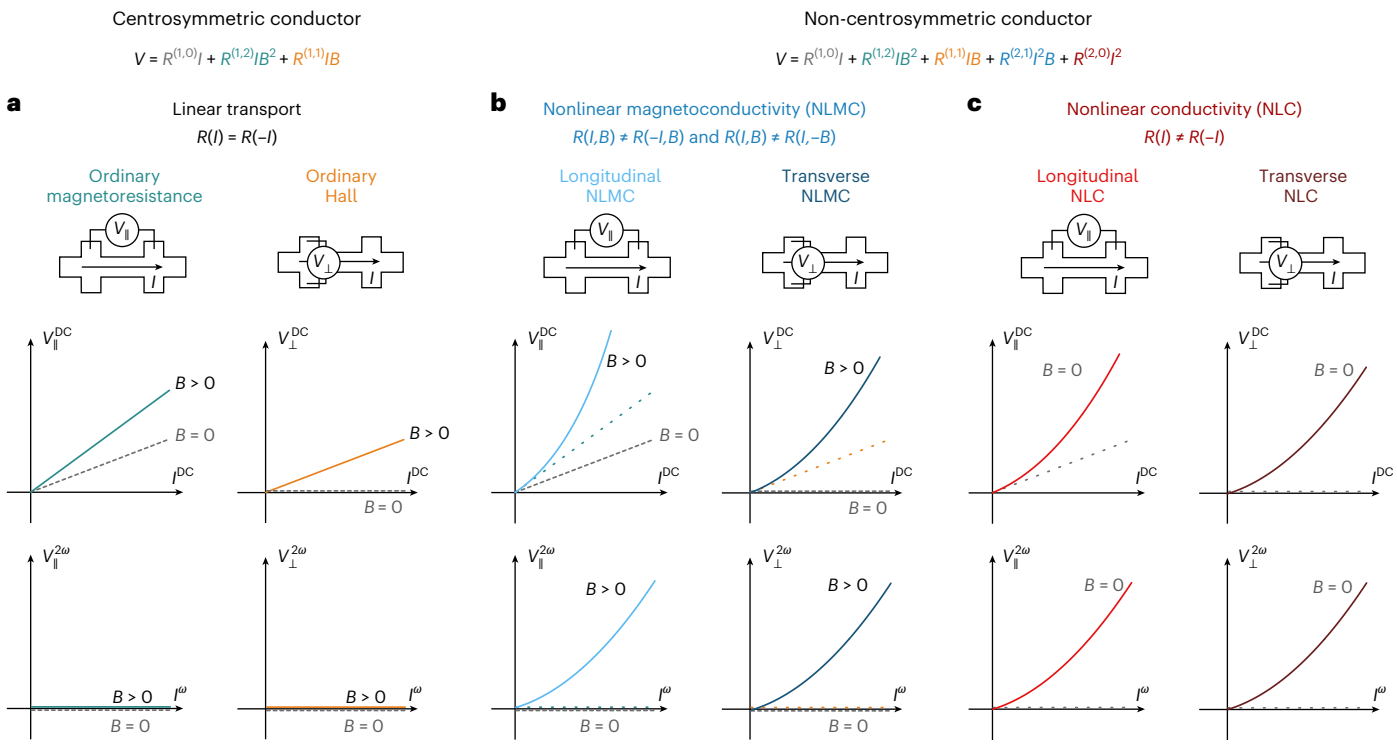

Fig. 1 | Different transport components in centrosymmetric and non-centrosymmetric conductors. a–**c**, Schematics of linear transport (**a**), NLMC (**b**) and NLC (**c**), highlighting the corresponding measurement configurations (longitudinal and transverse). The longitudinal and transverse DC voltage ($V_\parallel^{DC}$, $V_\perp^{DC}$; top row of plots) and second-harmonic voltage ($V_\parallel^{2\omega}$, $V_\perp^{2\omega}$; bottom row of plots) versus current (I^{DC}, I^{ω}) characteristics are also shown. When the overall signal (solid line) is nonlinear, the loosely dashed line represents its linear contribution. In the presence of an external magnetic field (B), the densely dashed line denotes the signal at $B = 0$. Importantly, in the second-harmonic voltage (bottom row), only the nonlinear contributions emerge. For clarity in this Review, we refer to bilinear magnetoresistance, unidirectional magnetoresistance and electrical magnetochiral anisotropy as longitudinal NLMC, and the nonlinear planar Hall effect as transverse NLMC. We also note that transverse NLC is often referred to as the nonlinear Hall effect (NLHE) in the literature, but this term should be reserved exclusively for true Hall responses.

material symmetry, the procedure for predicting both NLMC and NLC from the point group symmetry of a system will be explained. Moreover, one section will be devoted to discussing the physical microscopic mechanisms behind these nonlinear effects, outlining strategies to disentangle them in experiments. Finally, different applications of nonlinear transport effects will be explored, with a particular focus on spintronics and energy harvesting.

Non-centrosymmetric systems

Systems in which inversion symmetry ($\mathcal{P}$) is broken are called non-centrosymmetric. The inversion symmetry operation involves transforming a coordinate (x, y, z) into ($-x$, $-y$, $-z$). Therefore, in non-centrosymmetric systems, moving in one direction is not equivalent to moving in the opposite direction. This is exactly what happens when an electrical current is applied: electrons travelling in one direction experience different resistance from those travelling in the opposite direction, leading to nonlinear transport effects. In this section we will review the main reports on nonlinear transport effects in non-centrosymmetric systems, both in the presence and absence of an external magnetic field. Notably, the nonlinear responses across all configurations can be predicted using the point group of the material. Thus, we will begin by discussing this analysis, which will provide a clearer understanding of the experimental results presented later.

Crystal symmetry origin

Nonlinear transport effects are intrinsically linked to the symmetry of the system under study. As a result, the material's point group symmetry can be used to predict its nonlinear response. In centrosymmetric point groups, nonlinear transport is forbidden. However, once $\mathcal{P}$ is broken, nonlinear transport can occur in specific configurations. The number and nature of these configurations depend on which additional symmetry elements are broken. Whereas a measured voltage in response to an applied current is the magnitude used experimentally, it is more convenient to use a current density (j) in response to an applied electric field (E) for the theoretical description of the phenomenon. Below, we present the configurations of j, E and B for which the nonlinear transport is allowed in each point group (Supplementary Table 1) and discuss the procedure for linking them to the experimental results.

The second-order nonlinear current density ($j_i^{2\omega}$) in response to an electric field ($E_{j,k}^{\omega}$) can be expressed through the material's NLC tensor ($\sigma_{ijk}^{(2,0)}$) as: $j_i^{2\omega} = \sigma_{ijk}^{(2,0)} E_j^{\omega} E_k^{\omega}$. This tensor is known for each point group and can be found in the Bilbao Crystallographic Server under the name of nonlinear susceptibility[43]. Although some physical mechanisms are exclusive to the Hall response (see 'Nonlinear conductivity'), the tensor incorporates both ohmic and Hall responses[44], along with longitudinal and transverse components (see 'Longitudinal and transverse versus ohmic (dissipative) and Hall (non-dissipative), in Box 1), underscoring the suitability of using the common term NLC. As mentioned above, although the theoretical description is derived in terms of the NLC ($\sigma_{ijk}^{(2,0)}$), experimentally, it is useful to express the equation in terms of the nonlinear resistivity ($\rho_{ijk}^{(2,0)}$), $E_i^{2\omega} = \rho_{ijk}^{(2,0)} j_j^{\omega} j_k^{\omega}$. By inverting the matrix, the relationship between nonlinear resistivity and conductivity is obtained: $\rho_{ijk}^{(2,0)} = -\rho_{il}^{(1,0)} \sigma_{lmn}^{(2,0)} \rho_{mj}^{(1,0)} \rho_{nk}^{(1,0)}$, where $\rho_{ij}^{(1,0)}$ denotes the ohmic resistivity ($E_i^{\omega} = \rho_{ij}^{(1,0)} j_j^{\omega}$). Therefore, the second-order electric field ($E_i^{2\omega}$) in response to a current density ($j_{j,k}^{\omega}$) can be written as: $E_i^{2\omega} = -\rho_{il}^{(1,0)} \sigma_{lmn}^{(2,0)} \rho_{mj}^{(1,0)} \rho_{nk}^{(1,0)} j_j^{\omega} j_k^{\omega}$ (ref. 45). Note that non-

centrosymmetric systems generally exhibit anisotropic resistivity. Thus, as in the linear regime (see 'Longitudinal and transverse versus ohmic (dissipative) and Hall (non-dissipative), in Box 1), when currents are applied off-axis, nonlinear longitudinal tensor components may emerge in a transverse configuration and vice versa[32]. Moreover, in triclinic and monoclinic systems ($\rho_{ij}^{(1,0)} \neq 0$ with $i \neq j$), even when voltages are measured in response to currents along the principal axes, both longitudinal and transverse components may arise in both measurement configurations.

For effects induced by a magnetic field, the same procedure is followed. The second-order nonlinear current density ($j_i^{2\omega}$) in response to an electric field ($E_{j,k}^{\omega}$) and a magnetic field (B_l) can be written in terms of the NLMC tensor ($\sigma_{ijkl}^{(2,1)}$) as: $j_i^{2\omega} = \sigma_{ijkl}^{(2,1)} E_j^{\omega} E_k^{\omega} B_l$. The NLMC tensor is not directly available in the Bilbao Crystallographic Server database but can be generated using the Jahn symbol eV[V^2]V (ref. 46). As above, it is convenient to write the equation in terms of the nonlinear magnetoresistivity ($\rho_{ijkl}^{(2,1)}$), $E_i^{2\omega} = \rho_{ijkl}^{(2,1)} j_j^{\omega} j_k^{\omega} B_l$. By inverting the matrix, the NLMC contribution to the second-order electric field ($E_i^{2\omega}$) in response to a current density ($j_{j,k}^{\omega}$) and a magnetic field (B_l) is obtained: $E_i^{2\omega} = -\rho_{im}^{(1,0)} \sigma_{mnql}^{(2,1)} \rho_{nj}^{(1,0)} \rho_{qk}^{(1,0)} j_j^{\omega} j_k^{\omega} B_l$ (ref. 45). Furthermore, an additional contribution arising from a combination of the NLC and the ordinary Hall effect is also possible, but it occurs at a higher order in resistivity[45].

In Supplementary Table 1, we summarize the j, E and B configurations for which the nonlinear transport is allowed in each non-magnetic point group, both in the presence and absence of an external magnetic field. For the NLMC, the required direction of B is also indicated. Extra components may be allowed when E_j^{ω} is applied between two principal axes ($j_i^{2\omega} \propto E_j^{\omega} E_k^{\omega}$, with $j \neq k$), but we focus on the cases when E_j^{ω} is along a principal axis because they are the most studied configurations in experiments. Therefore, by referring to Supplementary Table 1, it is possible to directly predict the nonlinear components in any non-centrosymmetric system with a known point group. In addition, this enables verification of whether the observed nonlinear signals align with the material's symmetry. If they do not, surface contributions, thermal effects or other artefacts may be experimentally important.

Experimental observation

Polar systems. These systems exhibit structural asymmetry characterized by a well-defined directionality, which includes two or more points that remain unmoved under every symmetry operation. Depending on the number of unmoved points, they can define a line, a plane or all of space (polar direction). These systems usually host electric polarization, which must lie parallel to the polar direction. Spontaneous and modulable electric polarizations result in ferroelectric materials. Notably, the modulation of electric polarization affects the symmetry of the material and, therefore, it will have an impact on the nonlinear transport.

In polar WTe_2 (Fig. 2a), transverse NLC was observed and associated with the purely transverse Berry curvature dipole (BCD) mechanism, demonstrating a Hall effect in a time-reversal-invariant system and quickly attracting notable attention[30,31] (Fig. 2b). However, extrinsic mechanisms were also found to be important[31]. Shortly after, also in WTe_2, ferroelectric control of the nonlinear signal was demonstrated, showing that the NLC can be exploited to detect the electric polarization orientation[47]. The NLC has been reported in other polar transition metal dichalcogenides (TMDCs), such as $MoTe_2$ (ref. 48), and in out-of-plane configurations, with the voltage recorded perpendicular to the TMDC plane[49]. Notably, the effect was observed in samples grown via chemical vapour deposition, highlighting its potential for real-world applications[48]. Non-TMDC polar materials have also shown NLC with different microscopic mechanisms. For instance, in $NbIrTe_4$ (ref. 50), it is dominated by the BCD, whereas in BiTeBr (ref. 51) and $BaMnSb_2$ (ref. 52), extrinsic mechanisms are the major contribution.

In WTe_2, NLMC was also reported, in both longitudinal[19,53,54] (Fig. 2c) and transverse[25,54] configurations. The dependence of the crystallographic direction was studied[19,53], yielding results compatible with its polar symmetry (Supplementary Table 1). Indeed, the NLMC signal is consistently observed when the magnetic field is perpendicular to the direction of voltage detection. Similar results were obtained in the polar bulk semiconductor BiTeBr (ref. 16). The symmetry of the nonlinear signal was associated with the helical spin texture of these systems, showing that NLMC can be leveraged for detecting the spin texture (for more details, see 'Spintronics and orbitronics').

Chiral systems. The term 'chiral' refers to an object that produces a non-superimposable mirror image of itself, with human hands being the classical example. More precisely, a chiral system lacks any symmetry operation that reverses orientation (mirror reflection, inversion or rotoinversion) and therefore cannot be mapped onto its mirror image through rotations and translations alone. Most of the recent experimental reports on nonlinear transport in chiral materials have focused on elemental tellurium (Te). This crystal is formed by covalently bonded Te helices, stacked together through van der Waals interactions (Fig. 2d). Furthermore, it is a naturally p-doped semiconductor that presents good electronic-transport properties, and its handedness can be identified via transmission electron microscopy[55].

After NLMC was reported in artificially made bismuth (Bi) helices[14], it was demonstrated in a pristine system—macroscopic Te crystals[15]. However, the observed signal did not align with Te chiral symmetry. Later studies using Te flakes revealed the predicted behaviour[23], highlighting the critical role of symmetry analysis in ruling out potential artefacts (Supplementary Table 1). In addition, gate tunability was shown, and a sign change between right- and left-handed flakes was detected[23] (Fig. 2e). These findings were confirmed in subsequent experiments[56]. Recently, the crystallographic direction dependence of NLMC was studied in both parallel and transverse configurations, perfectly matching the expected symmetry for Te (ref. 45). In contrast to polar systems, NLMC is observed when the magnetic field is parallel to the direction of voltage detection, aligning with the radial spin texture of chiral systems[57] (for more details, see 'Spintronics and orbitronics'). In the chiral helimagnet $CrNb_3S_6$, an NLMC signal was reported, which is enhanced below the magnetic ordering temperature, illustrating the influence of chiral magnetic order on the nonlinear transport[58]. Furthermore, NLMC was observed in a chiral molecular conductor, confirming the chiral character of charge transport in molecular materials[59].

In the absence of an external magnetic field, longitudinal NLC was reported in Te and was demonstrated to be odd under spatial inversion[32] (Fig. 2f). In agreement with the NLMC measurements[45], the resistivity dependence of the nonlinear signal suggests that extrinsic scattering from dynamic sources is the dominant microscopic mechanism[32]. Cheng et al. reported the transverse NLC effect in Te (ref. 60); however, the observed signal does not align with Te symmetry (Supplementary Table 1, point group 32). These authors propose that surface symmetry breaking explains the discrepancy[60].

Surfaces and interfaces. In non-centrosymmetric crystals, $\mathcal{P}$ is broken in the bulk. However, all systems break $\mathcal{P}$ at the surface level. Therefore, if electronic transport is dominated by surface states, such as in two-dimensional electron gases (2DEGs) or topological insulators (TIs), nonlinear phenomena may arise even when $\mathcal{P}$ is preserved in the bulk[61].

The NLMC has been widely observed in 2DEGs[18,20,22,24] and TIs[17,62,63]. In these systems, $\mathcal{P}$ is broken at the surface level, while rotation axis and mirror planes around the surface normal direction are generally preserved. Under these conditions, and similar to polar systems, helical spin textures emerge and NLMC signals are recorded with in-plane magnetic fields perpendicular to the current[20,22,62] (for more details, see 'Spintronics and orbitronics'). When rotation axes and mirror

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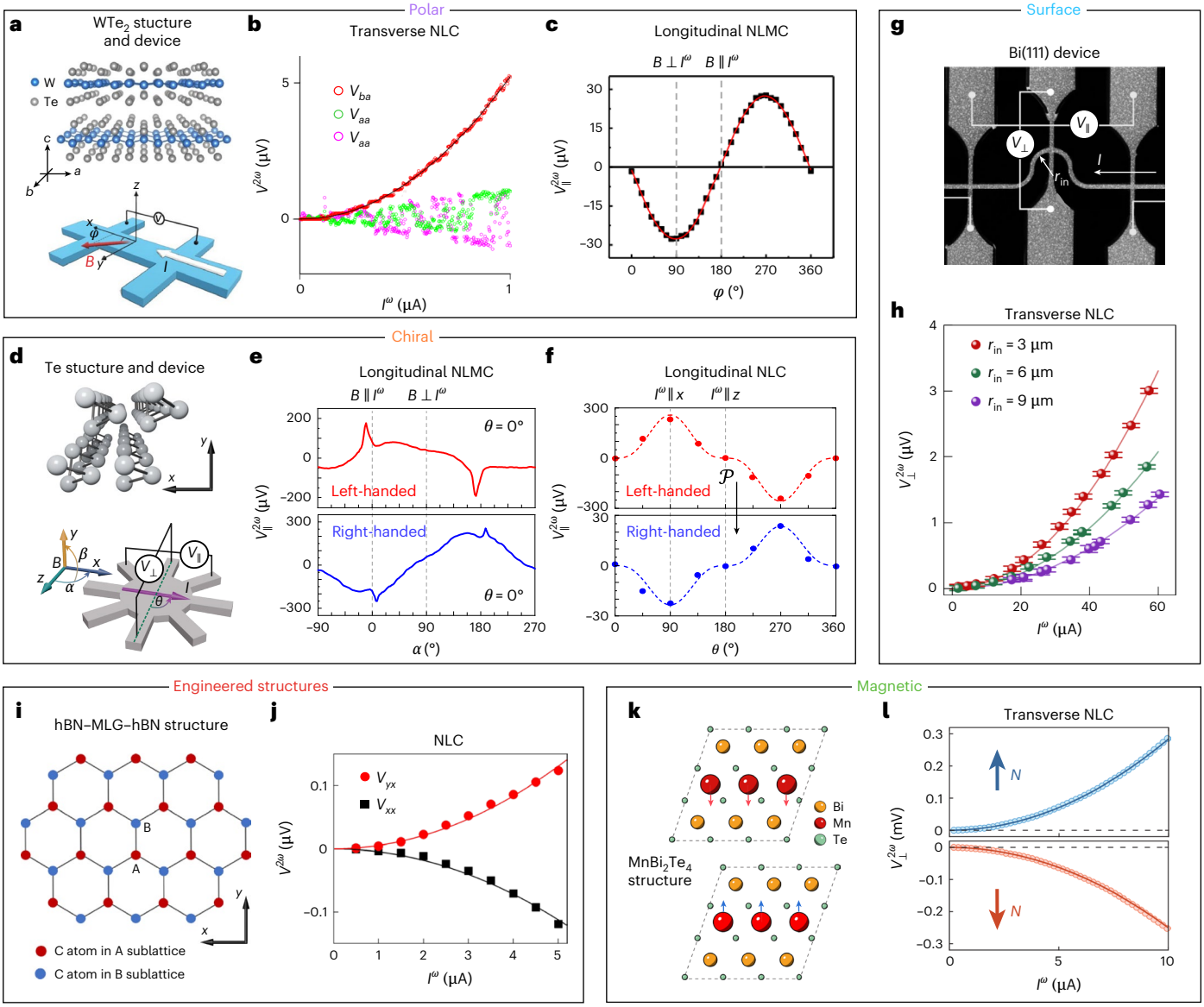

Fig. 2 | Nonlinear transport in non-centrosymmetric materials. **a**, Crystal structure of layered WTe_2 (top) and device geometry (bottom). **b**, Second-harmonic voltage ($V^{2\omega}$) as a function of the current (I^{ω}) in WTe_2. The nonlinear signal is only observed in the transverse configuration. **c**, $V^{2\omega}$ as a function of the in-plane angle φ between an external magnetic field (B) and I^{ω}, using the device shown in **a**. The signal is maximum when B and I^{ω} are mutually perpendicular, in agreement with WTe_2 polar symmetry. $V^{2\omega}$ is detected parallel to I^{ω}. **d**, Crystal structure of elemental Te (top) and device geometry (bottom). **e**, $V^{2\omega}$ as a function of the in-plane angle α between B and I^{ω} along the chiral z axis, using the device shown in **d**. The maximum signal is recorded when B and I^{ω} are parallel, in agreement with Te chiral symmetry. **f**, $V^{2\omega}$ as a function of the angle θ between I^{ω} and the chiral z axis, using the device shown in **d**. The signal is maximum when I^{ω} is along the x axis, in agreement with Te symmetry. In **e** and **f**, $V^{2\omega}$ is detected in the parallel configuration and changes sign between left- and right-handed Te crystals. **g**, Bi(111) device geometry. r_{in}, inner radius. **h**, Transverse $V^{2\omega}$ as a function of I^{ω} in Bi at room temperature. The signal is enhanced using an arc-shaped geometry. **i**, Inequivalent carbon sublattices in the hBN–MLG–hBN heterostructure. **j**, $V^{2\omega}$ as a function of I^{ω} in hBN–MLG–hBN. Nonlinear signals are recorded in both longitudinal and transverse configurations. **k**, Lattice structure of $MnBi_2Te_4$. **l**, Transverse $V^{2\omega}$ as a function of I^{ω} in $MnBi_2Te_4$. The nonlinear signal changes sign by reversing the Néel vector (N). Panels adapted with permission from: **a**(bottom),**c**, ref. 53 under a Creative Commons licence CC BY 4.0; **b**, ref. 30, Springer Nature Limited; **g**,**h**, ref. 69, Springer Nature Limited; **i**,**j**, ref. 33, Springer Nature Limited; **k**, ref. 86, Springer Nature Limited; **l**, ref. 85, AAAS. Panel **a**(top) reproduced with permission from ref. 53 under a Creative Commons licence CC BY 4.0.

planes perpendicular to the surface are also broken, NLMC signals with out-of-plane magnetic fields are observed as well[17,18]. In general, the NLMC has been measured in the longitudinal configuration; however, it has also been reported in the transverse configuration in both TIs and 2DEGs[25]. Furthermore, in these systems, the ratio between surface and bulk transport can be controlled via electrostatic gating, leading to highly modulable NLMC signals[20,22,62].

NLC has been reported in 2DEGs[64], TIs[65] and Weyl semimetals[66], surviving up to room temperature[67]. Moreover, non-volatile ferroic switching of the nonlinear signal has been demonstrated in $Pb_{1-x}Sn_xTe$ (ref. 68). NLC has been observed at room temperature on the (111) surface of Bi, which is easily grown and compatible with CMOS (complementary metal–oxide–semiconductor) technology and flexible substrates[69] (Fig. 2g). The reported nonlinear signal, which is connected to the side-jump and skew-scattering mechanisms, can be enhanced using an arc-shaped geometry[70], offering a new strategy by which to manipulate nonlinear phenomena (Fig. 2h).

Engineered structures. Crystalline systems found in nature or grown using standard methods are often centrosymmetric, which restricts the

number of materials that are capable of hosting nonlinear transport effects. However, van der Waals systems offer a new alternative. These materials can be exfoliated into single or multiple atomic layers and then stretched or combined, giving rise to new systems with different physical properties. Moreover, the stacking of layers can be engineered (for example, by twisting) to break specific symmetry elements[71]. Therefore, non-centrosymmetric engineered structures provide an ideal platform for studying nonlinear transport effects[72].

Graphene was the first van der Waals material to be exfoliated down to a single layer, beginning a new era in materials science[73]. However, monolayer graphene (MLG) presents a centrosymmetric structure, which forbids nonlinear transport effects. Notably, graphene is generally capped or sandwiched by hexagonal boron nitride (hBN) to improve its electronic properties. The presence of hBN breaks $\mathcal{P}$, enabling nonlinear transport phenomena (Fig. 2i). The hBN/MLG/hBN heterostructure was the first system to demonstrate NLC in both longitudinal and transverse configurations within a single structure[33] (Fig. 2j). The longitudinal response cannot originate from the BCD, with skew scattering identified as the dominant microscopic mechanism. Recently, it was discovered that breaking $\mathcal{P}$ by twisting graphene layers provides the possibility of introducing completely new physics, such as superconductivity[74]. In the field of nonlinear transport, twisted bilayer graphene[34,75], twisted double bilayer graphene[76] and twisted trilayer graphene[77] have all reported NLC signals with giant outputs. In the absence of hBN or twisting, bilayer graphene can be artificially corrugated via lithographically patterned strain, breaking $\mathcal{P}$ and exhibiting NLC signals[78].

Other van der Waals systems, different from graphene, have also been engineered to break the required symmetries. For instance, stretching[79] or twisting[80] WSe_2 results in surprisingly high NLC signals. Moreover, a heterostructure formed by WSe_2 and WTe_2 has demonstrated ferroelectric control of BCD, giving rise to highly modulable nonlinear phenomena[81]. Other approaches, such as applying an additional electric field to generate a sizable BCD[82], have also been reported.

Magnetic systems. In non-centrosymmetric magnetic materials, besides $\mathcal{P}$, $\mathcal{T}$ is also broken. Therefore, even though they could be classified under previous sections, we discuss them separately. Indeed, in this case, nonlinear effects governed by the magnetization (M) or Néel vector (N) emerge. These additional contributions, where M or N plays a role similar to B, can be interpreted as an NLC that is dependent on the magnetic vectors but also as an anomalous version of NLMC, highlighting their distinct origin and $\mathcal{T}$-odd nature ($V^{2\omega} \propto I^2 M$ or $V^{2\omega} \propto I^2 N$).

$MnBi_2Te_4$ is a non-centrosymmetric TI in which the manganese (Mn) spins are aligned parallel in each layer but are coupled antiparallel to adjacent layers (Fig. 2k). Therefore, odd-layer $MnBi_2Te_4$ is a topological ferromagnet, whereas even-layer $MnBi_2Te_4$ is a topological antiferromagnet. On the one hand, odd-layer $MnBi_2Te_4$ presents chiral edge states, which lead to nonlinear output voltages with different signs at opposite edges[83]. Notably, by reversing the magnetization direction, the sign of the nonlinear output also changes. Similar results have been found in chromium-doped $(Bi,Sb)_2Te_3$ (ref. 84).

On the other hand, even-layer $MnBi_2Te_4$ has been reported to exhibit nonlinear transport in both transverse[85,86] and longitudinal[86,87] configurations. In these systems, both $\mathcal{P}$ and $\mathcal{T}$ symmetries are broken, but $\mathcal{PT}$ symmetry is preserved below the Néel temperature. Note that the emergence of $\mathcal{PT}$ symmetry cancels the BCD, so it cannot be the origin of the reported effects. The field-induced correction to Berry curvature, also called Berry connection polarizability (BCP) or positional shift correction, was proposed to give rise to a $\mathcal{PT}$-invariant nonlinear transport effect[88] and is believed to be the main mechanism behind these experiments. This mechanism is often quoted to be related to a band-normalized quantum metric (for more details, see 'Anomalous nonlinear magnetoconductivity'), and has gained much attention in the community. Remarkably, the reported signal can be modulated by reversing the Néel vector N (Fig. 2l). However, it is important to note that control of the nonlinear signal via manipulation of the Néel vector had previously been demonstrated in CuMnAs using current pulses[89]. More recently, manipulation with moderate magnetic fields has also been reported in Mn_3Sn, leading to highly tunable nonlinear phenomena[90]. Even-layer CrSBr is a non-centrosymmetric antiferromagnet at low fields that, with the application of moderate magnetic fields, undergoes a metamagnetic transition to a ferromagnetic state. In this system, anomalous NLMC has been reported, which is fully modulable by both the magnetization and Néel vectors[91]. All of these results demonstrate that the anomalous NLMC can be used to determine the orientation of the magnetization in ferromagnets and the Néel vector in antiferromagnets, providing opportunities for the development of electrical readout in magnetic devices based on nonlinear transport effects.

Finally, it is important to note that some systems present a centrosymmetric crystal structure but undergo phase transitions, for example, topological or magnetic phase transitions, breaking $\mathcal{P}$ and thus allowing nonlinear transport effects. Therefore, nonlinear transport measurements can be exploited to probe such transitions[92–94].

In Supplementary Table 2, the main experimental reports on nonlinear transport in non-centrosymmetric materials are summarized. The most critical parameters for comparing the results—second-harmonic output voltage ($V^{2\omega}$), applied current (I^{ω}), temperature (T) range, ohmic resistivity ($\rho^{(1,0)}$) and magnetic field (B) (for NLMC)—are included. In addition, the NLC ($\sigma^{(2,0)}$), nonlinear resistivity ($\rho^{(2,0)}$), NLMC ($\sigma^{(2,1)}$) and nonlinear magnetoresistivity ($\rho^{(2,1)}$) were calculated. Both $\sigma^{(2,0)}$ and $\sigma^{(2,1)}$ offer insight into the magnitude of the physical effect itself, whereas $\rho^{(2,0)}$ and $\rho^{(2,1)}$ indicate the system's efficiency in generating a large $V^{2\omega}$. Consequently, $\rho^{(2,0)}$ and $\rho^{(2,1)}$ serve as the figure of merit for applications such as energy harvesting, where maximizing $V^{2\omega}$ is critical (see 'Energy harvesting'). From these parameters, we can derive different strategies for enhancing the nonlinear output (see 'Outlook').

Physical mechanisms

Beyond their significance from a material symmetry perspective, nonlinear effects have enabled the experimental observation of previously hidden physical mechanisms. In this section we summarize the different microscopic mechanisms identified for each nonlinear transport effect, providing insights into how to disentangle them in experiments.

Nonlinear conductivity

Berry curvature dipole. The intrinsic contribution to the conventional linear Hall effect is the integral of the Berry curvature (Ω) over the Fermi sea. In a $\mathcal{T}$-invariant system, the Berry curvature is odd in momentum space $\Omega(k) = -\Omega(-k)$, so its integral, weighted by the equilibrium Fermi distribution (f_0), must vanish because the states at k and $-k$ are equally occupied. However, a second-order response arises from the integral of the Berry curvature over the non-equilibrium distribution of electrons. As this non-equilibrium distribution is unequal for k and $-k$ states, the integral can be finite. Seminal studies on nonlinear transport effects in non-centrosymmetric systems date back to 1968[95]. More recently, Deyo et al. developed a semiclassical theory highlighting the contributions of Berry curvature and side-jump mechanisms[96]. Sodemann and Fu[97], by generalizing previous work on photocurrents by Moore and Orenstein[98], demonstrated that the intrinsic NLHE effect is indeed tied to the integral of the Berry curvature gradient over the occupied states, $D_{ab} = \int_k f_0 \left(\partial_a \Omega_b\right)$, known as the BCD. Thus, the key intrinsic property that is responsible for the non-dissipative NLC (NLHE) in non-magnetic crystals with broken $\mathcal{P}$ is the BCD. Note that, in the direct current (DC) limit, the BCD contribution of the NLC tensor is linearly proportional to the scattering time (τ), $\sigma^{(2,0)} \propto \tau D$ (Table 1). We note that, for some authors, intrinsic contributions are necessarily scattering-independent, and thus the BCD is considered an extrinsic

Table 1 | Nonlinear transport mechanisms

Mechanism	BCD	SK	SJ	BC–ZC	BCP	NLD	ASK
Origin	Intrinsic	Extrinsic	Extrinsic	Intrinsic	Intrinsic	Intrinsic	Extrinsic
NLC	✓	✓	✓	×	×	×	×
NLMC	×	✓	✓	✓	✓	×	×
Anomalous NLMC	×	✓	✓	×	✓	✓	✓
Longitudinal	×	✓	✓	✓	✓	✓	×
Transverse	✓	✓	✓	✓	✓	✓	✓
C_{3v} symmetry	×	✓	✓	✓	✓	✓	✓
$\mathcal{P}$ symmetry	×	×	×	×	×	×	×
$\mathcal{T}$ symmetry	✓	✓	✓	×	×	×	×
$\mathcal{PT}$ symmetry	×	×	×	×	✓	✓	✓
τ-scaling	τ^1	$\tau^{0,1,2,3,4}$	$\tau^{0,1,2,3,4}$	τ^2	τ^0	τ^2	τ^2

The origin, contribution to NLC, NLMC and anomalous NLMC, measurement configurations permitted (longitudinal or transverse), symmetry constraints (C_{3v}, $\mathcal{P}$, $\mathcal{T}$, $\mathcal{PT}$) and τ-scaling are shown for each microscopic mechanism: SK, skew-scattering; SJ, side-jump; BC–ZC, Berry curvature and Zeeman coupling; NLD, nonlinear Drude; and ASK, anomalous skew-scattering. SK and SJ mechanisms contribute with multiple τ-scalings to the nonlinear transport signal ($\tau^{0,1,2,3,4}$). Under $\mathcal{T}$ conditions, the SK mechanism contributes just with $\tau^{1,2,3}$ and the SJ mechanism with $\tau^{1,2}$.

mechanism[99]. However, in the framework adopted in this Review, extrinsic refers to mechanisms where scattering centres have been explicitly incorporated into the calculations.

Several studies[97,100,101] have highlighted monolayer TMDCs, such as WTe_2 and $MoTe_2$, as promising candidates for experimentally observing the NLHE. However, $\mathcal{P}$ and/or C_{3v} symmetry in these systems cancels the effect. Therefore, symmetry must be broken externally, for instance, by applying an out-of-plane electric field or through the presence of an interacting substrate. On the other hand, even-layer WTe_2 naturally breaks $\mathcal{P}$ in its pristine state. Ma et al.[30] and Kang et al.[31] reported the observation of the NLHE in this system. Whereas the results have been explained in terms of the BCD, signatures of extrinsic contributions have also been detected[31].

It should be noted that strong BCD is not limited to noncentrosymmetric TMDCs, as shown in 'Experimental observation'. The BCD is greatly enhanced and changes sign during topological transitions[93], making these transitions experimentally observable[92]. This phenomenon provides opportunities for the development of Berry-curvature-based memory devices.

Extrinsic mechanisms and scaling law. The study of intrinsic and extrinsic mechanisms of the linear anomalous Hall effect has evolved over the years due to the complexity of experimentally disentangling the various contributions[102]. After numerous proposals, a refined scaling law for the anomalous Hall effect was established[103], identifying Berry curvature as the intrinsic mechanism, whereas skew-scattering and side-jump mechanisms were recognized as the extrinsic mechanisms. This understanding was applied to develop a scaling law for NLC soon after its experimental observation[104,105]. As discussed in the previous section, the BCD serves as the intrinsic mechanism. By analogy with the anomalous Hall effect, skew-scattering and side-jump mechanisms have been proposed as extrinsic mechanisms[106]. Notably, whereas the BCD induces exclusively Hall signals, the extrinsic mechanisms also lead to ohmic[44] and longitudinal[32,33] NLC (Table 1). Moreover, nonlinear transport is highly dependent on the crystallographic direction (see 'Crystal symmetry origin'). Indeed, a quantum theory has been developed that identifies components of the NLC tensor as purely extrinsic for certain space groups[107]. Therefore, by studying the dependence of the crystallographic direction and the scaling law of the nonlinear response, it may be possible to disentangle the different contributions in experiments.

Nonlinear magnetoconductivity

Berry curvature and Zeeman coupling. Liu et al. used Berry-curvature-corrected semiclassical and Boltzmann formalism to develop a microscopic theory of NLMC[46]. They recognize both Berry curvature and Zeeman intrinsic terms, which scale quadratically with τ. The Zeeman term can be further separated into spin and orbital contributions, with the former requiring strong spin–orbit coupling (SOC) in the system, but not the latter. Therefore, NLMC can be utilized to explore the spin and orbital contributions within a system. The proposed theory has been applied to chiral elemental Te, where discrepancies between theory and experiment have been observed, highlighting the importance of considering extrinsic mechanisms as well[45]. Notably, the Zeeman and Berry curvature terms apply to both longitudinal and transverse effects, which is crucial for disentangling them from other contributions via experiments.

Berry connection polarizability. The NLMC can be understood as an NLC in a system perturbed by B. By analogy to the intrinsic linear anomalous Hall effect, the anomalous velocity of electrons is given by $v^A = E \times \Omega$, but to study the nonlinear response at the second-order E^2, the electric field correction of the Berry curvature (Ω^E) should be considered, that is, $v^A = E \times \Omega^E$. This correction was calculated by Gao et al.[88], showing that $\Omega^E = \nabla_k \times \mathcal{A}^E$, where $\mathcal{A}^E_a = \widetilde{G_{ab}} E_b$ is the field-corrected Berry connection and $\widetilde{G_{ab}}$ is the positional shift or BCP. Note that, here, the BCP is defined with respect to the B-perturbed band structure. By expressing all quantities in the original unperturbed eigenstates, Huang et al.[108] demonstrated that the nonlinear response is given by the integral over the occupied states of two products: the band velocity multiplied by the spin susceptibility of BCP and the k-space dipole of BCP multiplied by the spin magnetic moment. Later results also suggested an orbital contribution[109]. Note that the B-perturbed BCP contribution to the NLMC ($\sigma^{(2,1)}_{ijkl}$) is reported to be τ-independent and purely transverse[108,109]. However, regarding the pure transversality, there are some discrepancies within the community (for more details, see 'Berry connection polarizability and nonlinear Drude').

Extrinsic mechanisms and scaling law. Experiments have highlighted the significance of extrinsic contributions to the NLMC. In this regard, a scaling law has been developed for $\mathcal{T}$-odd nonlinear transport in magnetic materials[110]. In these systems, the internal magnetic vectors play a role similar to that of B in the NLMC. However, B induces additional orbital magnetic field contributions. To address this difference and obtain a proper scaling law for NLMC, an experimental study has modified the scaling law for $\mathcal{T}$-odd nonlinear transport in magnetic systems including the additional orbital magnetic field contributions. By applying the proposed scaling law, the dominant microscopic mechanisms were identified for the case of chiral Te (ref. 45).

Anomalous nonlinear magnetoconductivity

As discussed in 'Magnetic systems', non-centrosymmetric magnetic materials, besides breaking $\mathcal{P}$, also break $\mathcal{T}$, giving rise to $\mathcal{T}$-odd mechanisms governed by the magnetization or Néel vectors. Notably, there are systems that break both $\mathcal{P}$ and $\mathcal{T}$, but not $\mathcal{PT}$ symmetry. The preservation of $\mathcal{PT}$ symmetry suppresses most of the mechanisms discussed in previous sections, such as the BCD, skew-scattering and side-jump mechanisms (Table 1). Therefore, systems that preserve $\mathcal{PT}$ symmetry while breaking both $\mathcal{P}$ and $\mathcal{T}$ are of special interest for studying the mechanisms covered in this section.

Berry connection polarizability and nonlinear Drude. Whereas we have already discussed the role of the BCP in the NLMC, we will now review that, in magnetic systems, the BCP contributes to nonlinear transport even in the absence of B. As detailed in 'Berry connection polarizability', the Berry connection receives a correction from the applied electric field $\Omega^E = \nabla_k \times \mathcal{A}^E$, where $\mathcal{A}^E_a = G_{ab}E_b$. Here, as we are not considering any B, the BCP (G_{ab}) is defined relative to the unperturbed band structure. Following the same strategy used for the NLMC, Liu et al.[99] and Wang et al.[111] calculated the τ-independent contribution to the nonlinear transverse transport in magnetic systems, showing that it stems from the integral over the occupied states of the BCP dipole moment. Subsequent studies based on the quantum response theory have suggested that this τ-independent term may also play a role in longitudinal transport[112,113]. This hypothesis was substantiated by a recent investigation that explicitly demonstrated the equivalence between semiclassical and quantum response theories up to second order in the electric field[114].

We want to highlight that this τ-independent mechanism has been referred to in the literature as the band-normalized quantum metric[85,86,112,113]. The quantum metric and Berry curvature represent the real and imaginary parts of the quantum geometric tensor, respectively. In its definition, the BCP contains an extra energy denominator compared with the quantum metric, which generically makes it a different, non-geometric quantity. Whereas there may be an approximate connection between the two when only two bands are involved, the BCP correction remains a more faithful name to the meaning of this quantity, and we have thus used it in this way in this Review. This is, of course, not withstanding the legitimate high interest of the quantum metric in condensed matter systems and general quantum transport, which may be found to be connected in other ways in the future.

In addition, a nonlinear Drude term has also been reported[112,115]. This term originates from the skewness of the energy dispersion, which is generally non-zero when both $\mathcal{P}$ and $\mathcal{T}$ are separately broken. It depends only on the band dispersion of the states at the Fermi surface, and on the scattering time τ. Whereas the linear Drude term is proportional to τ, the nonlinear Drude is proportional to τ^2, which makes it $\mathcal{T}$-odd but $\mathcal{PT}$-even. The nonlinear Drude term in magnetic systems is equivalent to the Zeeman coupling term in the NLMC (see 'Berry curvature and Zeeman coupling').

Extrinsic mechanism and scaling law. $\mathcal{PT}$ symmetry cancels out the Berry curvature and the conventional skew-scattering mechanisms discussed in the previous sections[116]. However, in magnetic systems, other extrinsic mechanisms may still emerge, even under conditions of $\mathcal{PT}$ symmetry. For example, an anomalous skew-scattering mechanism has been proposed, where Berry curvature and skew scattering work in tandem[116]. A scaling law for $\mathcal{T}$-odd nonlinear transport in magnetic systems (anomalous NLMC) has been proposed[110]. However, non-centrosymmetric magnetic systems generally also exhibit a broad range of mechanisms that contribute to $\mathcal{T}$-even transport (NLC), resulting in numerous possibilities. Therefore, whereas several scaling laws have been proposed[110,115,117], it is crucial also to consider the different symmetry constraints of these mechanisms (Table 1) to disentangle them via experiments.

Applications

Besides their importance in accessing intriguing physical properties of studied systems, nonlinear transport effects have several important applications.

Spintronics and orbitronics

Spintronics and orbitronics are branches of electronics that harness, respectively, the spin and orbital angular momentum of electrons in addition to their charge. They promise considerably lower power consumption[118,119], although, to expand their applications, it is crucial to develop systems that enable accurate generation, control and detection of electron angular momentum. In this regard, non-centrosymmetric systems may offer an outstanding platform.

In systems with broken $\mathcal{P}$, SOC has traditionally been considered necessary for band splitting, leading to the formation of spin (S) textures[120,121], $E_S(k) \neq E_{-S}(k)$. However, recent studies have suggested that, in the absence of SOC, orbital angular momentum (L) can similarly lift band degeneracy, generating orbital textures instead[122,123], $E_L(k) \neq E_{-L}(k)$. Edelstein predicted that, when an electric field is applied, these textures force transport electrons to align their angular momentum in a specific direction, resulting in the accumulation of angular momentum[124]. This accumulation can be intuitively understood as a current-induced internal magnetic field. Consequently, an external magnetic field parallel or antiparallel to the current-induced internal magnetic field will lead to different effective fields, causing a resistance difference for opposite current directions (Fig. 3a). This phenomenon is none other than the NLMC and provides a simple method by which to probe angular momentum textures through purely electrical means[17,20].

Besides this intuitive interpretation, two detailed theories have been developed to explain the longitudinal NLMC in terms of the spin texture of TIs[17,21] and 2DEGs[20]. The first approach is based on the conversion of a non-equilibrium spin current into a charge current under the application of an external magnetic field[17], whereas the second focuses on the interplay between current-induced spin polarization and scattering processes due to inhomogeneities of spin-momentum locking[20,21]. The former has also been generalized to explain the transverse NLMC[25]. Whereas these microscopic descriptions offer an alternative approach to explaining NLMC, the mechanisms discussed in 'Nonlinear magnetoconductivity' provide a more direct approach (Fig. 3b). Nonetheless, angular momentum textures exist and have been observed directly using spin-resolved angle-resolved photoemission spectroscopy[57,125]. Therefore, both strategies are valid, and by combining them, it is possible to exploit symmetry analysis to predict angular momentum textures and use NLMC measurements to probe them.

In particular, when $\mathcal{P}$ is broken while mirror symmetry is preserved, spin textures perpendicular to the mirror planes emerge (Fig. 3c). This is the classic example of 2DEGs[18,20,22] and TIs[17,62,63]. In these systems, when surface symmetry is maintained, two perpendicular mirror planes constrain the spins to the surface, resulting in helical spin textures[20,22,62]. However, if surface symmetry is disrupted and only one mirror plane is retained, out-of-plane spin components emerge[17,18]. Notably, as shown in Supplementary Table 1, polar groups with two mirror planes ($mm2$, $4mm$ and $6mm$) enable NLMC in only one configuration with the magnetic field aligned perpendicular to the current and both mirror planes, consistent with a helical spin texture. Conversely, polar groups with a single mirror plane (m and $3m$) allow NLMC in two configurations with the magnetic field oriented in both directions perpendicular to the current, also in agreement with the spin texture of those systems. Furthermore, when all mirror symmetries are broken, radial spin textures arise (Fig. 3c), as seen in chiral systems such as trigonal Te (ref. 57). Indeed, chiral groups permit NLMC with magnetic fields parallel to the current (Supplementary Table 1), aligning with a radial spin texture[23]. Therefore, the same symmetry conditions that give rise to angular momentum textures also allow NLMC effects. This

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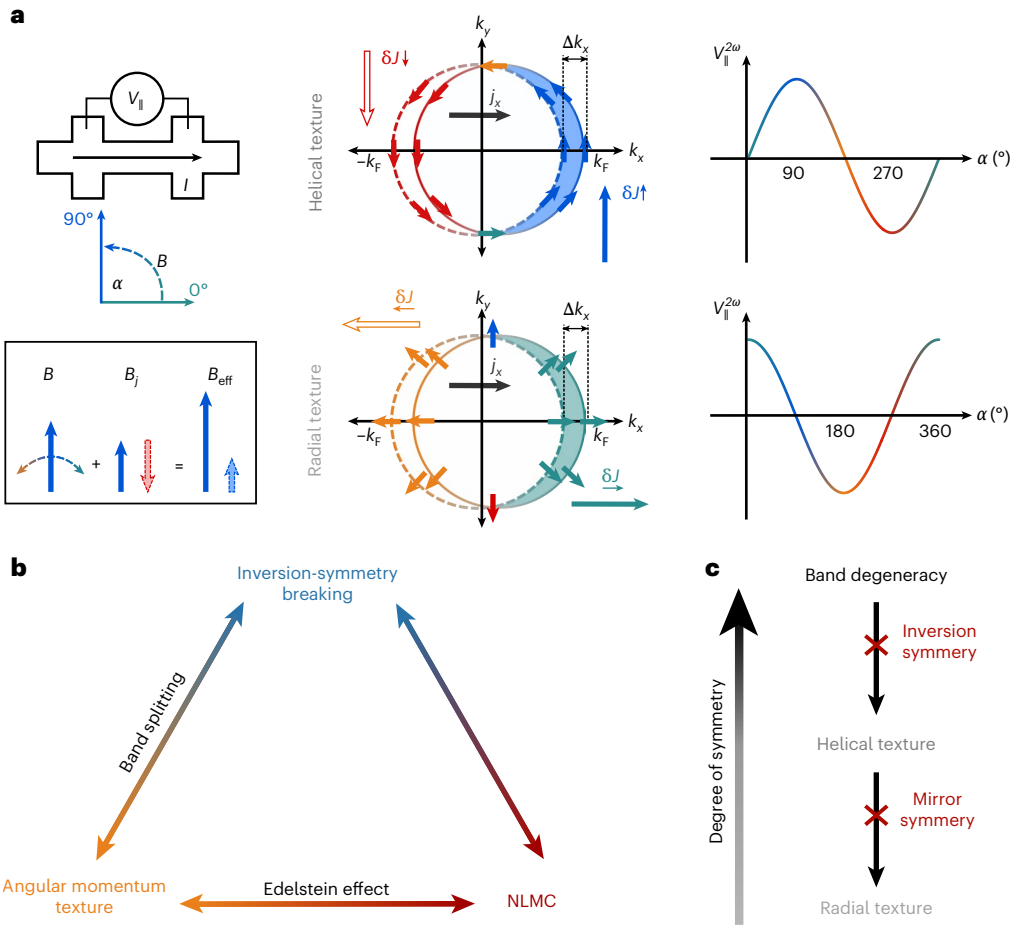

Fig. 3 | Connection between NLMC and angular momentum textures. a, Left: measurement configuration for probing helical and radial textures. The inset (bottom) shows the effective field (B_{eff}) as a result of the external field (B) and current-induced field (B_j). Middle: representation of both texture types, including the accumulation of current-induced angular momentum (δJ) in response to a current (j_x) through the Edelstein effect. k_F, Fermi wavevector; k_x and k_y, wavevector components along x and y, respectively; Δk_x, wavevector shift along x. Right: the expected dependence of the second-harmonic parallel voltage ($V_{\parallel}^{2\omega}$) on the angle α for each texture. **b**, Schematic showing the relationship between $\mathcal{P}$ breaking, the angular momentum texture and NLMC. **c**, Schematic showing the different symmetries that must be broken to allow helical and radial textures to be displayed.

interplay can be harnessed to probe angular momentum textures through nonlinear transport measurements. Moreover, following the symmetry analysis, it is possible to engineer systems with specific angular momentum textures[126], providing versatile platforms for applications in spintronics and orbitronics.

In this section, another physical effect should be mentioned, possibly related to NLMC, which is the so-called chirality-induced spin selectivity. This has recently attracted much attention as a way to generate large spin polarization in chiral molecular systems. However, the phenomenology is largely constrained to materials with very low electrical conductivity, and the main mechanisms behind the observed experiments remain unclear[127]. A theory has recently been proposed to explain how the spin accumulation generated by the spin Hall effect can also induce NLMC[128].

Energy harvesting

In recent years, there has been an exponential increase in ambient radiofrequency (RF) radiation, accompanied by the rapid proliferation of 'Internet of Things' (or IoT) devices[129]. Consequently, wireless rectifying antennas (rectennas), which can effectively harvest ambient RF radiation to power IoT devices, have emerged as a critical technology for reducing global energy consumption[130]. State-of-the-art rectennas combine an antenna to collect the RF radiation and convert it into an alternating current (AC) current with a rectifier that converts the AC signal into a DC current. Schottky diodes are the most widely used alternative for the rectification module. However, they exhibit a threshold input voltage, referred to as thermal voltage limitation, below which the device fails to operate effectively[35,131]. This makes the inclusion of an antenna module to enhance the incoming signal essential, leading to a substantial increase in the overall size of the rectenna and consequently constraining its applicability in microscale devices. In this context, non-centrosymmetric systems may offer an exceptional solution.

As discussed throughout the manuscript, non-centrosymmetric systems exhibit intrinsic nonlinear transport characteristics. Therefore, when an incoming RF wave (E^{in}) is absorbed by the material, an asymmetric electric field (E^{rec}) is generated. The temporal integration of E^{rec} results in a net non-zero DC current (I^{DC}) within the material[35,131] (Fig. 4a). Note that the physics of the rectification mechanism is completely different from that of Schottky diodes (for more details, see 'Physical mechanisms') and is therefore not restricted by the thermal voltage limitation. In low-power applications, this enables the elimination of the antenna module, dramatically reducing the dimensions of the rectenna to the microscale or even nanoscale.

The practical implementation of RF energy harvesting using a non-centrosymmetric system was demonstrated in $TaIrTe_4$, where symmetry is disrupted at the surface level[67]. As the rectification is based on the transverse NLC, V^{DC} is recorded transverse to the

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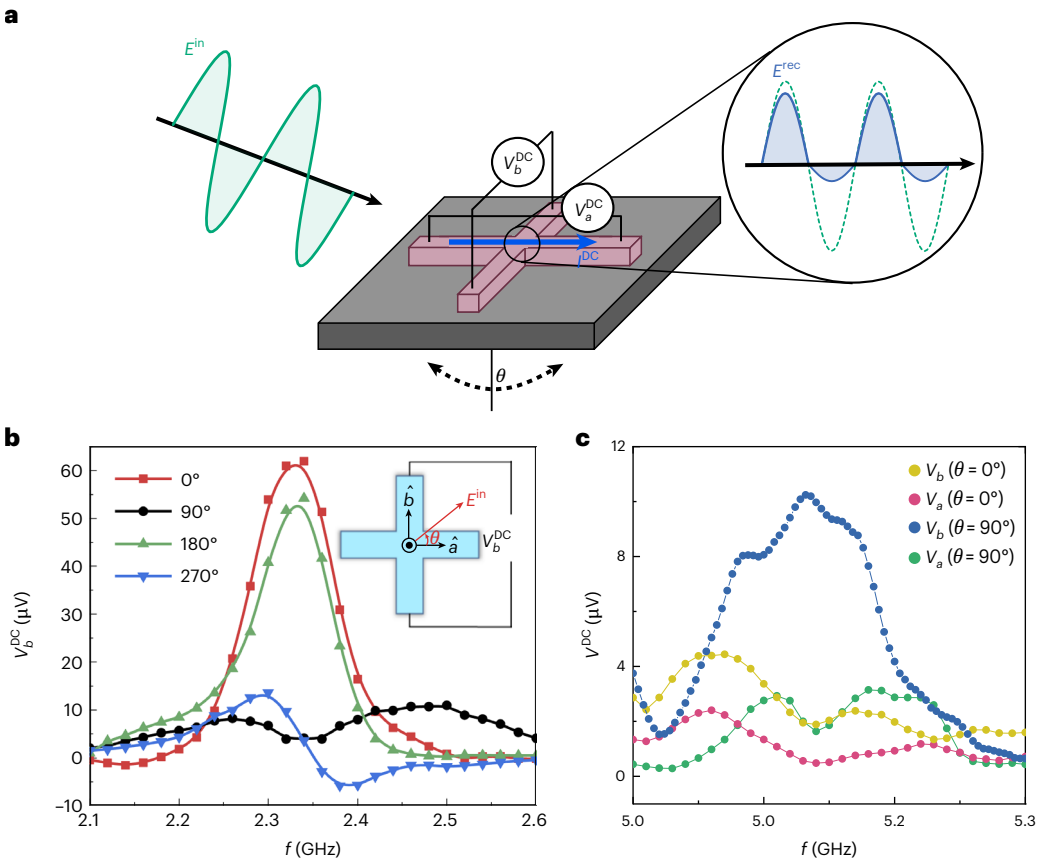

Fig. 4 | Wireless RF rectification. a, An incident RF wave (E^{in}) generates an asymmetric electric field (E^{rec}) due to the nonlinear transport characteristics of non-centrosymmetric systems, resulting in a DC current (I^{DC}) within the material. **b**,**c**, Rectified V^{DC} as a function of frequency (f) in $TaIrTe_4$ (**b**) and Te (**c**). The signal is at a maximum when the incident polarization is transverse (**b**) (parallel (**c**)) to the direction of voltage detection, in agreement with the nonlinear transport characteristics of $TaIrTe_4$ (**b**) (Te (**c**)). The inset in **b** shows the relative orientation between E^{in} and the a and b axes, defined by the angle θ. Panels adapted with permission from: **b**, ref. 67, Springer Nature Limited; **c**, ref. 132, Wiley.

Table 2 | Summary of the main experimental results on wireless RF rectification

Material	Mechanism	T (K)	f (GHz)	P^{in} (nW)	P^{out} (nW)	η (%)	Ref.
$TaIrTe_4$	BCD, extrinsic	300	2.45	0.9	0.02	0.023	67
$MnBi_2Te_4$	BCP	20	0.2–20	—	0.01	—	85
BiTeBr	Extrinsic	300	0.2–5.9	—	0.53	—	51
Te	Extrinsic	10–300	5.15	0.95	0.04	0.04	132
WTe_2	BCD	300	2.45	3	0.0015	0.0005	133
Bi_2Se_3	Extrinsic	300	2.45	15.2	0.18	0.01	133
Bi_2Te_3	Extrinsic	300	2.45–27.4	3.18	0.47	0.15	133

Included here are the non-centrosymmetric material in which the effect is reported, the mechanism, temperature (T), frequency (f), incident power (P^{in}), output power (P^{out}) and efficiency ($\eta = P^{out}/P^{in}$).

polarization direction of E^{in} (Fig. 4b). Subsequent experiments in magnetic[85] and polar[51] systems also demonstrated wireless rectification in a transverse configuration. In addition, chiral Te exhibited RF rectification in a longitudinal configuration, with V^{DC} measured parallel to the polarization direction of E^{in}, consistent with its longitudinal NLC[132] (Fig. 4c). Recently, Bi_2Te_3 has exhibited rectification in both longitudinal and transverse configurations at frequencies as high as 27.4 GHz, which is crucial for the next generation of 5G frequency bands[133]. In Table 2, the main reports in literature are summarized. Notably, in all of these experiments, an antenna module to collect and enhance the incoming signal was not required, achieving a full energy harvesting performance at microscale dimensions. Although the efficiencies remain orders of magnitude below those of state-of-the-art devices that combine an antenna with a Schottky diode[132], these results pave the way for the development of self-powered microscale devices (see 'Outlook').

At higher frequencies, in the visible regime, an equivalent effect known as the bulk photovoltaic effect has also been observed. State-of-the-art photovoltaic cells are based on p–n junctions, whose efficiencies are limited by the Shockley–Queisser limit. As the bulk photovoltaic effect is not subject to this limit, it has attracted considerable attention for self-powered harvesting applications[134]. Although the underlying principle of the effect is the same as wireless RF rectification and is subject to the same symmetry constraints (Supplementary Table 1), the higher frequencies and energies of visible light result in electronic band transitions, different transport regimes, additional

microscopic mechanisms and thermal effects. Therefore, it falls outside the scope of this Review[135,136].

Outlook

In conclusion, the field of nonlinear transport effects in non-centrosymmetric materials is advancing rapidly, driven by both fundamental insights and its broad application potential. Despite considerable progress, key challenges and opportunities remain to be addressed.

First, for application purposes, there is growing interest in achieving stronger nonlinear responses. Intrinsic mechanisms, such as the BCD[97] and BCP[108], are well understood, enabling the identification of key material characteristics that enhance nonlinear transport. Although non-centrosymmetric materials for electronic transport are limited, materials engineering has shown promise in overcoming this limitation. Among the various approaches, van der Waals heterostructures, which offer the flexibility to stack[33] and twist[34] different layers, stand out as an ideal platform[137]. At the same time, experimental studies have revealed the pronounced and often dominant role of extrinsic mechanisms[32], which remain less understood. Advancing theoretical and experimental research in this area could be crucial for achieving substantial enhancements in nonlinear output.

Second, whereas initial experiments have primarily focused on output enhancements, non-centrosymmetric systems offer a broad range of possibilities. One particularly intriguing avenue is the exploration of non-volatile responses for memory applications. Nonlinear transport effects enable the direct reading of magnetization, Néel and polarization vector orientations[99], opening new pathways for device functionality. Preliminary studies on ferromagnetic[91], antiferromagnetic[85,86,89,91] and ferroelectric[68] systems have already demonstrated reconfigurable and permanent nonlinear responses, yet considerable potential remains to be explored in this field.

Third, wireless rectification has been demonstrated at room temperature[67], but its efficiency still falls behind that of state-of-the-art Schottky diode devices. The relationship between NLC and nonlinear resistivity shows that higher rectifier resistance leads to stronger nonlinear output voltages (see 'Experimental observation'). However, it is important to note that wireless rectification devices must transfer the harvested power[138]. This transfer process limits the rectifier resistance, as it becomes inefficient if the receiving device is more conductive[132]. To overcome this limitation, the mechanism behind rectification in non-centrosymmetric systems may also provide the solution, as it is not constrained by the bandgap. This enables the engineering of an optimal balance between resistivity and high absorption, facilitating both a strong rectification output and efficient power transfer. Moreover, the working frequency spectrum is not a fundamental limitation, meaning that the devices can be optimized to operate across a broad spectrum, from radiofrequency to far-infrared regions, including visible and terahertz frequencies.

Fourth, in the fields of spintronics and orbitronics, achieving magnetic states writing and reading with high precision and low power consumption is critical. In this regard, non-centrosymmetric materials present a promising alternative, as their symmetry can be engineered to drive and detect the targeted angular momentum orientations. This Review provides a framework for identifying such symmetries, which may guide the development of the next generation of spin–orbit torque and magneto-electric spin–orbit devices. Indeed, early applications of non-centrosymmetric systems in these devices have shown promising performance[126,139].

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Competing interests

The authors declare no competing interests.

Additional information

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